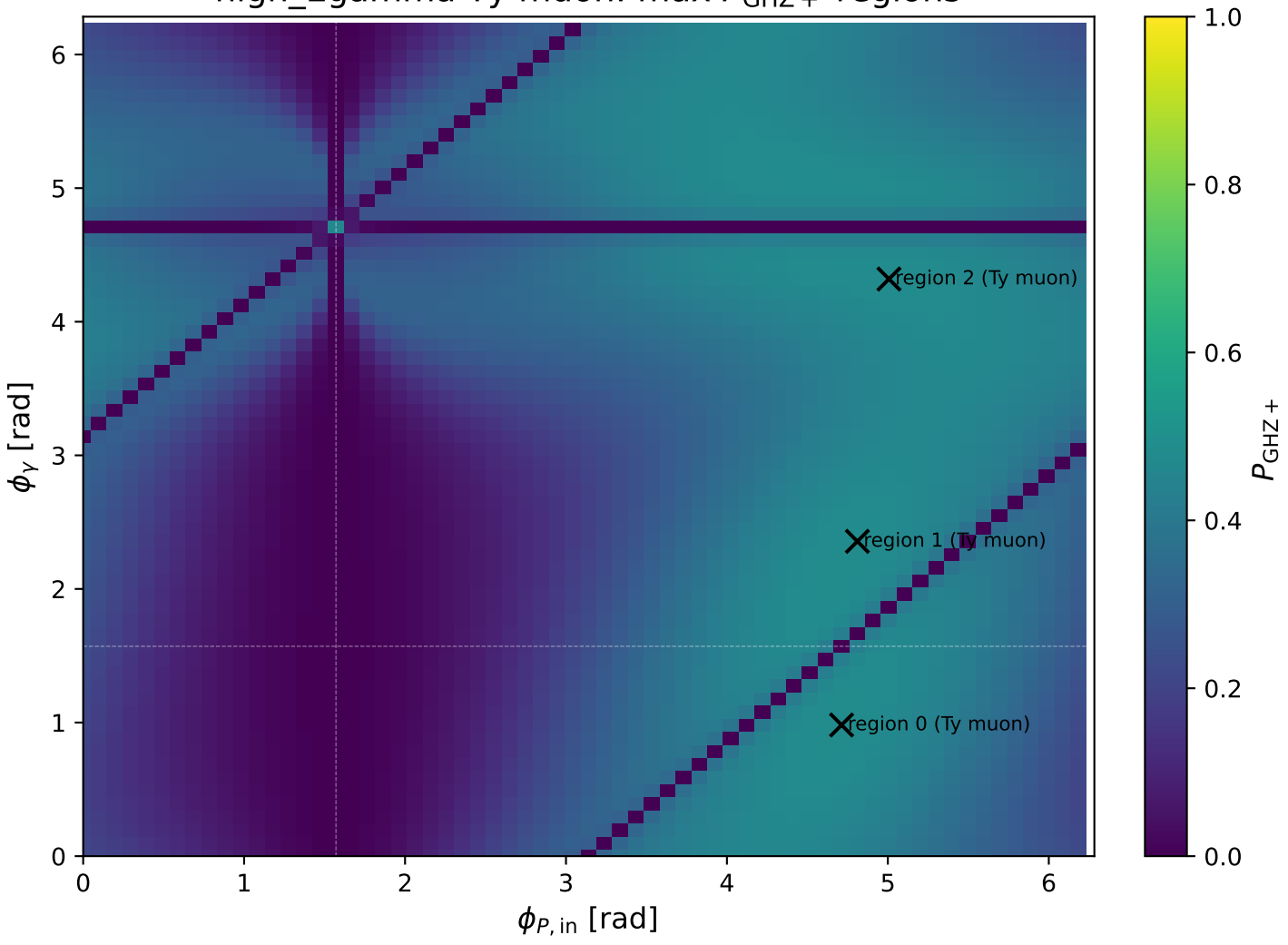
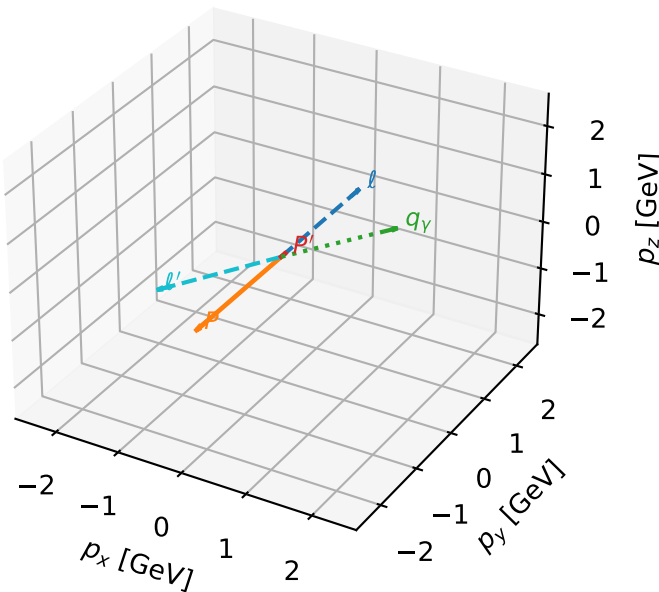


high\_Egamma Ty muon: max  $P_{\text{GHz}+}$  regions



# Ty muon characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

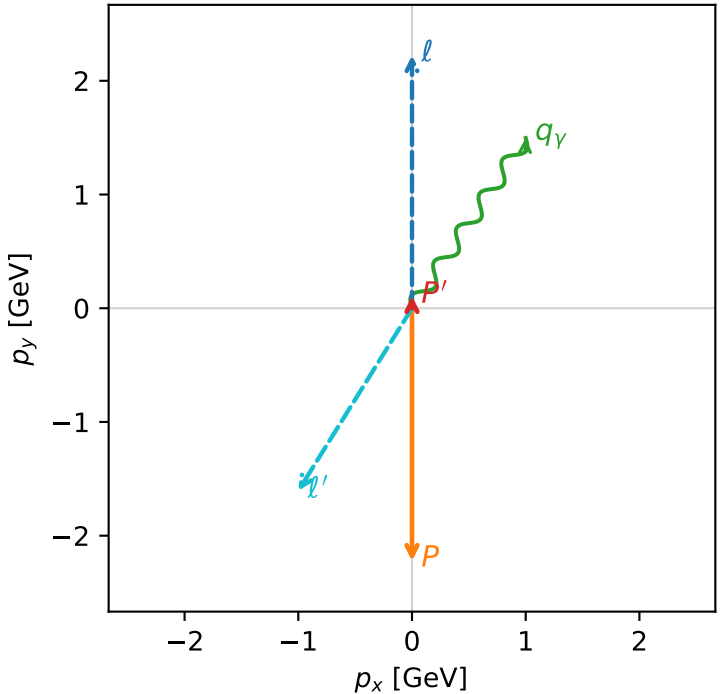


ghz\_purity\_high\_egamma\_ty\_electron\_region\_0  $P_{\text{GHZ}}=0.482446$   
region: high\_Egamma\_Ty\_electron\_region\_0  
outgoing-state purity: 0.50004  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.108644$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=0.981748$   
 $Q^2=15.5469$ ,  $x_B=0.834909$ ,  $t=-3.28022$ ,  $W^2=3.95402$ ,  $y=0.902847$   
 $\theta(l', \gamma)=178.171$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= 0.0000$   
 $l'$   $E= 1.8942$   $p_x= -1.0000$   $p_y= -1.6053$   $p_z= 0.0000$   
 $P'$   $E= 0.9443$   $p_x= 0.0000$   $p_y= 0.1086$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 1.0000$   $p_y= 1.4966$   $p_z= 0.0000$

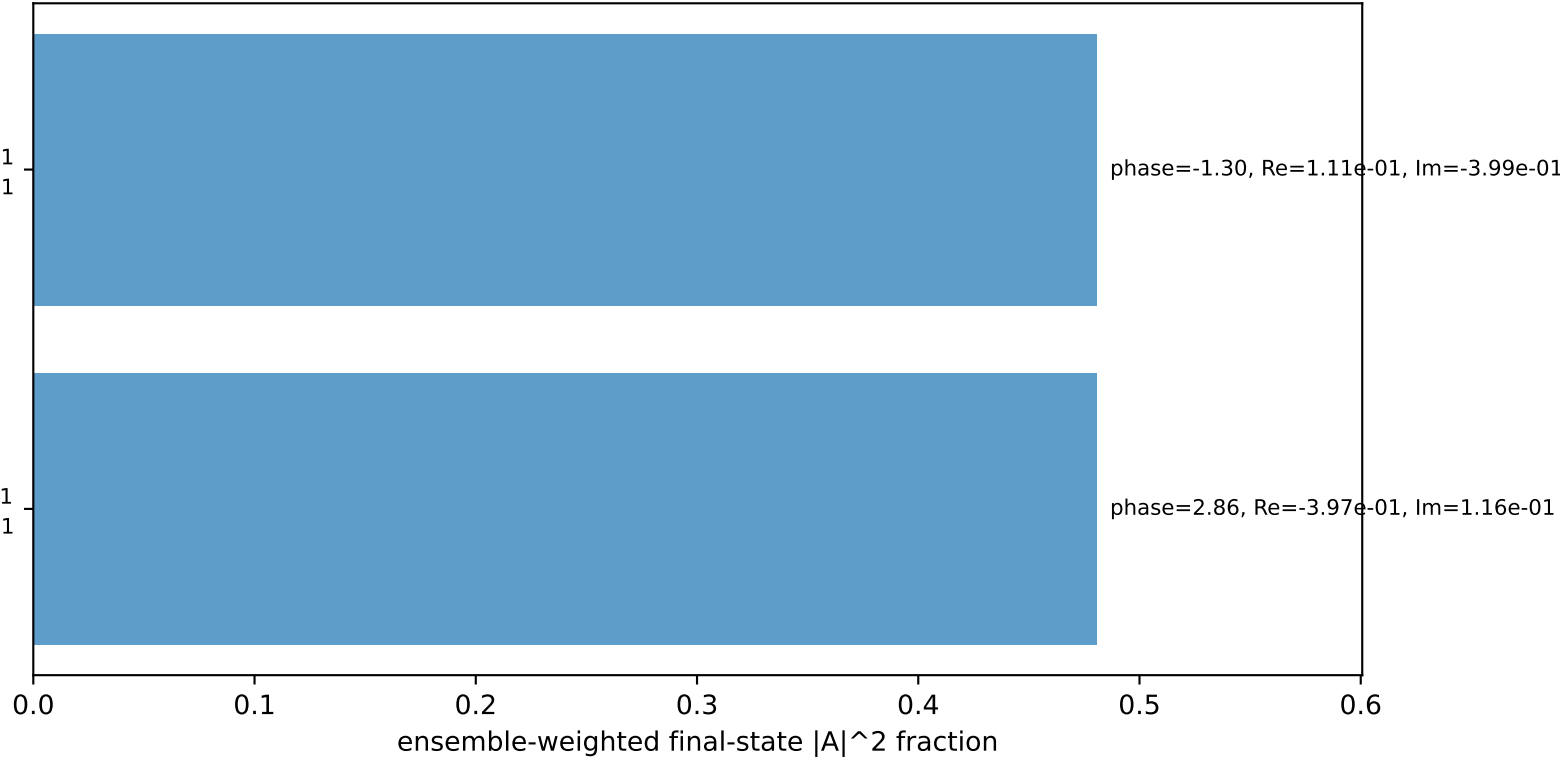
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
muon Ty, proton h=+1

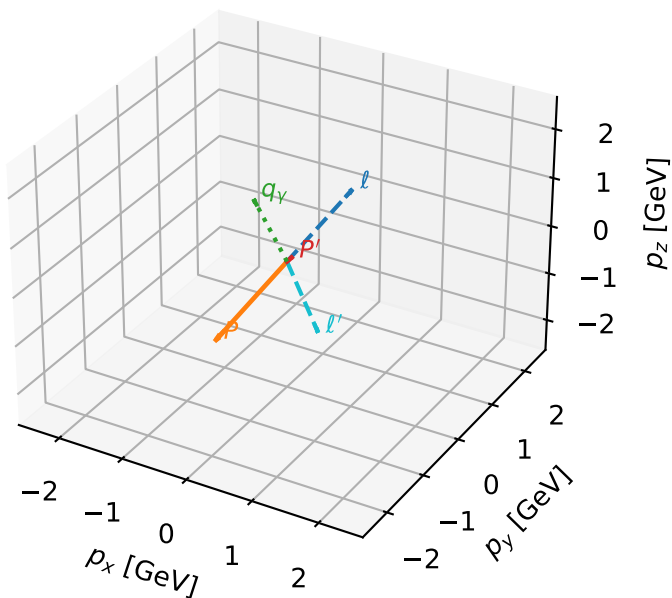
$h_e=-1, h_p=-1, h_\gamma=-1$   
muon Ty, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=96.1%)



# Ty muon characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

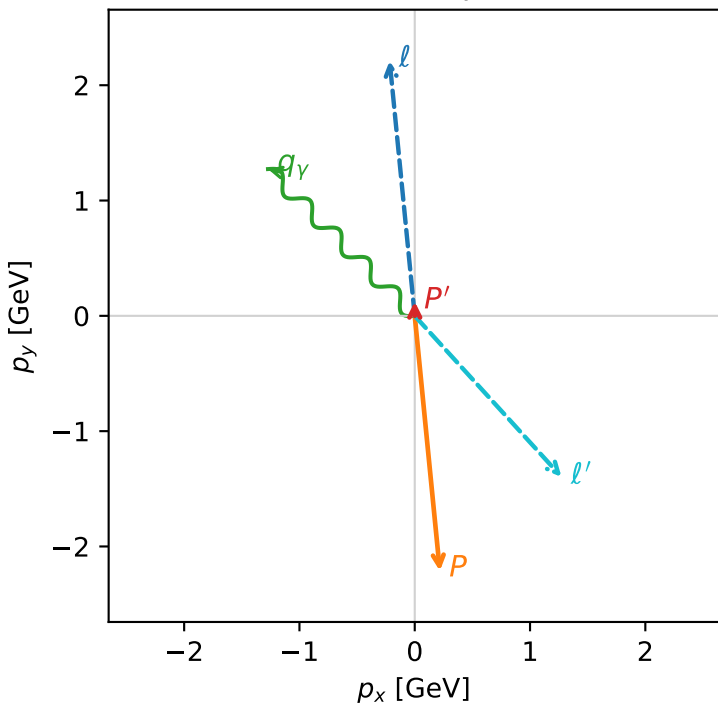


ghz\_purity\_high\_egamma\_ty\_electron\_region\_1  $P_{\{\text{rm GHZ}\}}=0.480795$   
 region: high\_Egamma\_Ty\_electron\_region\_1  
 outgoing-state purity: 0.500036  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.123642$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.66897$ ,  $\phi_P=4.81056$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=2.35619$   
 $Q^2=15.1349$ ,  $x_B=0.830394$ ,  $t=-3.35315$ ,  $W^2=3.97111$ ,  $y=0.883699$   
 $\theta(l', \gamma)=177.348$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -0.2179$   $p_y= 2.2124$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.2179$   $p_y= -2.2124$   $p_z= 0.0000$   
 $l'$   $E= 1.8924$   $p_x= 1.2728$   $p_y= -1.3964$   $p_z= 0.0000$   
 $P'$   $E= 0.9461$   $p_x= 0.0000$   $p_y= 0.1236$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.2728$   $p_y= 1.2728$   $p_z= 0.0000$

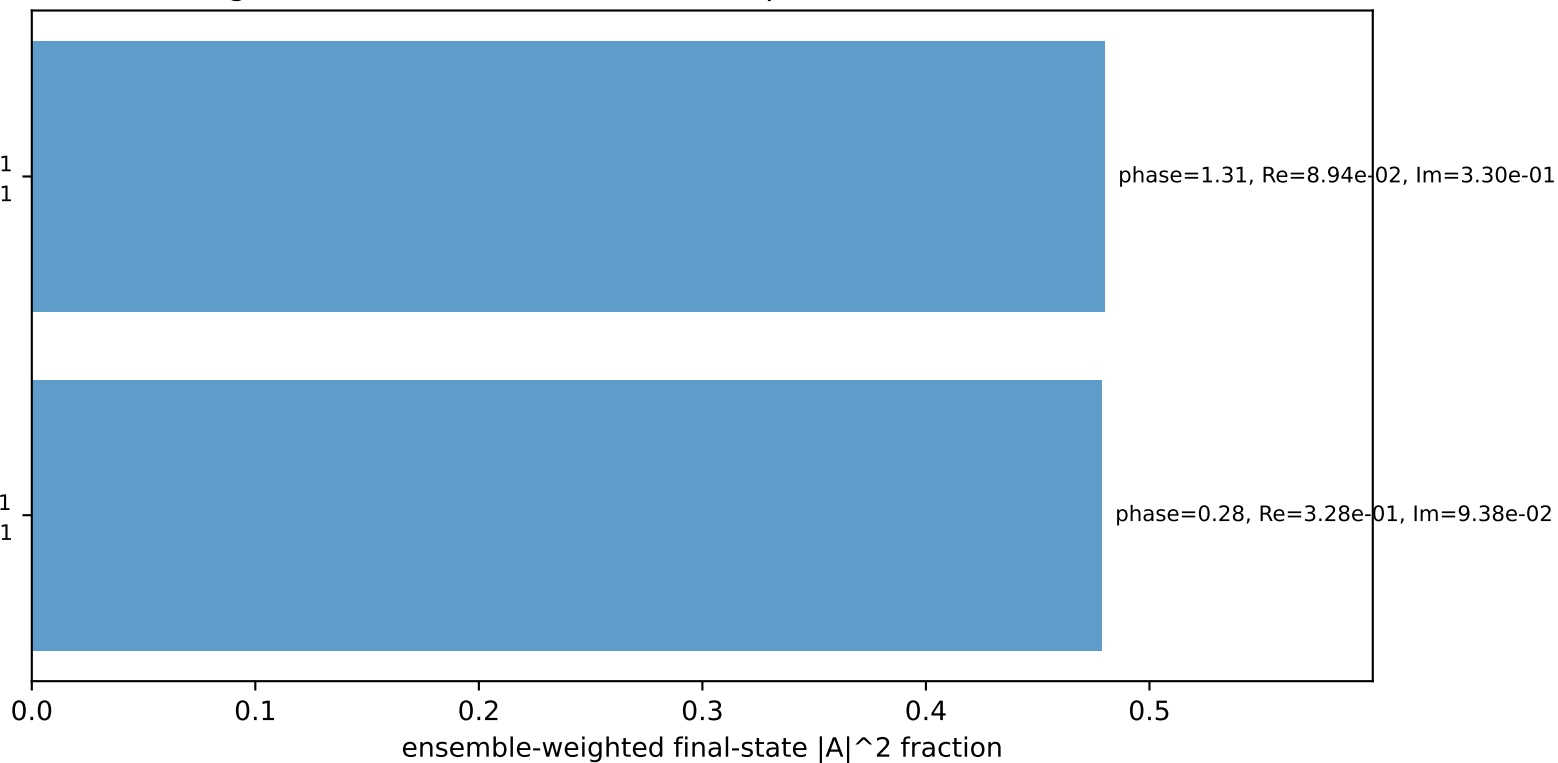
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
 muon Ty, proton h=+1

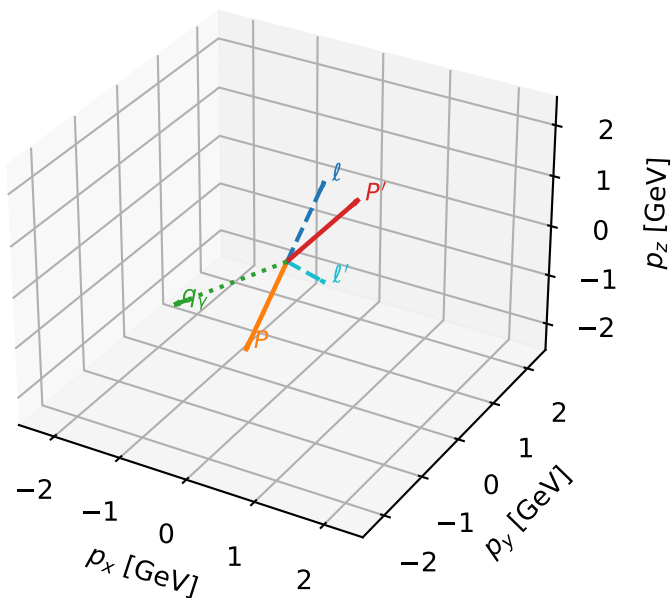
$h_e=-1, h_p=-1, h_\gamma=-1$   
 muon Ty, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=95.9%)



# Ty muon characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

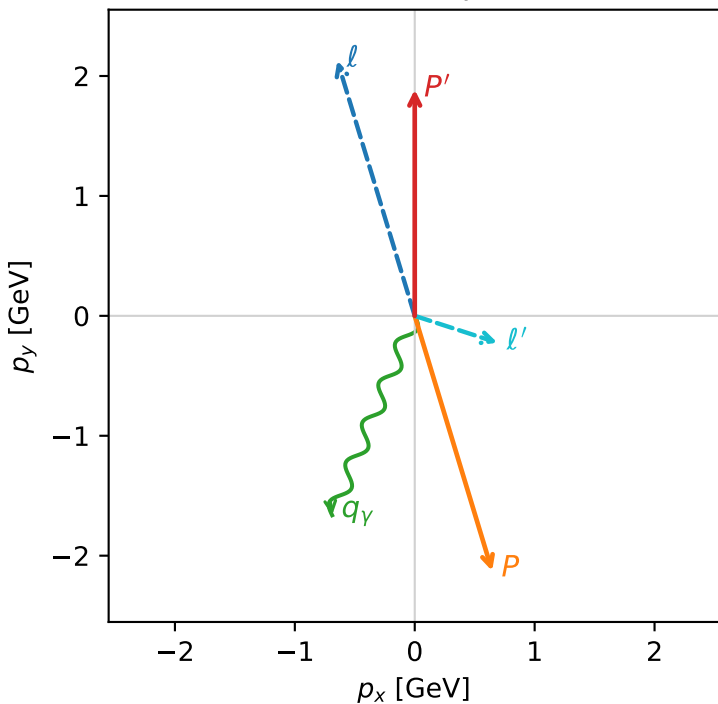


ghz\_purity\_high\_egamma\_ty\_electron\_region\_2  $P_{\text{GHZ}}=0.478826$   
 region: high\_Egamma\_Ty\_electron\_region\_2  
 outgoing-state purity: 0.500008  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,\mu}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=1.88636$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.86532$ ,  $\phi_P=5.00691$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=4.31969$   
 $Q^2=5.07463$ ,  $x_B=0.268034$ ,  $t=-16.4329$ ,  $W^2=14.738$ ,  $y=0.917958$   
 $\theta(l', \gamma)=94.5328$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -0.6453$   $p_y= 2.1274$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.6453$   $p_y= -2.1274$   $p_z= 0.0000$   
 $l'$   $E= 0.7318$   $p_x= 0.6888$   $p_y= -0.2234$   $p_z= 0.0000$   
 $P'$   $E= 2.1067$   $p_x= 0.0000$   $p_y= 1.8864$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -0.6888$   $p_y= -1.6630$   $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
 muon Ty, proton h=+1

$h_e=-1, h_p=-1, h_\gamma=-1$   
 muon Ty, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=95.7%)

